

1.5 The Dot Product

Important properties of dot product or scalar product:

- a) $\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}||\mathbf{B}| \cos \theta_{BA}$
- b) $\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}$
- c) work = $\int \mathbf{F} \cdot d\mathbf{r}$
- d) Electric Potential = $-\int \mathbf{E} \cdot d\mathbf{l}$
- e) $\mathbf{a}_x \cdot \mathbf{a}_y = \mathbf{a}_y \cdot \mathbf{a}_x = \mathbf{a}_x \cdot \mathbf{a}_z = \mathbf{a}_z \cdot \mathbf{a}_x = \mathbf{a}_y \cdot \mathbf{a}_z = \mathbf{a}_z \cdot \mathbf{a}_y = 0$
- f) $\mathbf{a}_x \cdot \mathbf{a}_x = \mathbf{a}_y \cdot \mathbf{a}_y = \mathbf{a}_z \cdot \mathbf{a}_z = 1$
- e) Resolution of vector in its horizontal and vertical components (Fig. 1.6).

$$A_x = A \cos \theta, \quad A_y = A \sin \theta$$

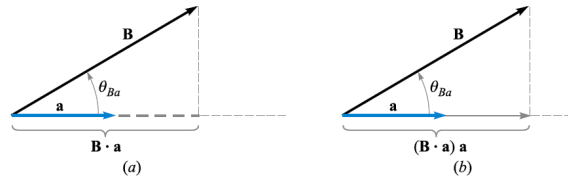


Fig. 1.6 (a) The scalar component of $\mathbf{B}$ in the direction of the unit vector $\mathbf{a}$ is $\mathbf{B} \cdot \mathbf{a}$ (b) The vector component of $\mathbf{B}$ in the direction of the unit vector $\mathbf{a}$ is $(\mathbf{B} \cdot \mathbf{a}) \mathbf{a}$

Example: Consider the vector field $\mathbf{G} = y\mathbf{a}_x - 2.5x\mathbf{a}_y + 3\mathbf{a}_z$ and the point $Q(4, 5, 2)$. We wish to find: $\mathbf{G}$ at Q ; the scalar component of $\mathbf{G}$ at Q in the direction of $\mathbf{a}_N = \frac{1}{3}(2\mathbf{a}_x + \mathbf{a}_y - 2\mathbf{a}_z)$; the vector component of $\mathbf{G}$ at Q in the direction of $\mathbf{a}_N$; and finally, the angle $\theta_{G\mathbf{a}_N}$ between $\mathbf{G}(\mathbf{r}_Q)$ and $\mathbf{a}_N$.

Solution: Substituting the coordinates of point Q into the expression for $\mathbf{G}$, we have

$$\mathbf{G}(\mathbf{r}_Q) = 5\mathbf{a}_x - 10\mathbf{a}_y + 3\mathbf{a}_z$$

Next we find the scalar component. Using the dot product, we have

$$\mathbf{G} \cdot \mathbf{a}_N = (5\mathbf{a}_x - 10\mathbf{a}_y + 3\mathbf{a}_z) \cdot \frac{1}{3}(2\mathbf{a}_x + \mathbf{a}_y - 2\mathbf{a}_z) = \frac{1}{3}(10 - 10 - 6) = -2$$

The vector component is obtained by multiplying the scalar component by the unit vector in the direction of $\mathbf{a}_N$,

$$(\mathbf{G} \cdot \mathbf{a}_N) \mathbf{a}_N = -(2) \frac{1}{3}(2\mathbf{a}_x + \mathbf{a}_y - 2\mathbf{a}_z) = -1.333\mathbf{a}_x - 0.667\mathbf{a}_y + 1.333\mathbf{a}_z$$

The angle between $\mathbf{G}(\mathbf{r}_Q)$ and $\mathbf{a}_N$ is found from

$$\begin{aligned}\mathbf{G} \cdot \mathbf{a}_N &= |\mathbf{G}| \cos \theta_{Ga} \\ -2 &= \sqrt{25 + 100 + 9} \cos \theta_{Ga}\end{aligned}$$

and

$$\theta_{Ga} = \cos^{-1} \frac{-2}{\sqrt{134}} = 99.9^\circ$$